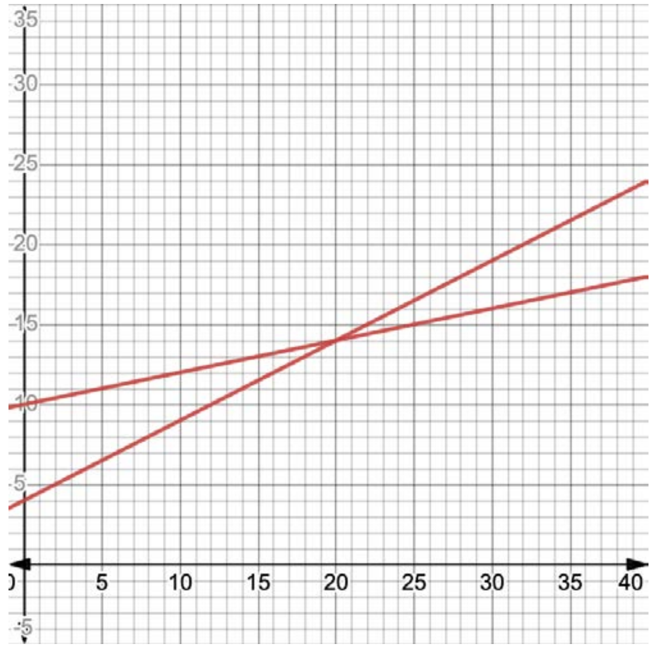
Algebra 1 | Name_Answer Key________
Unit 5 | Date____________________
Lesson 7 Practice | Period__________________

Use the system of equations $\begin{cases} y = 0.5x + 4 \\ y = 0.2x + 10 \end{cases}$ to complete the following. $\frac{2}{10} = \frac{1}{5}$



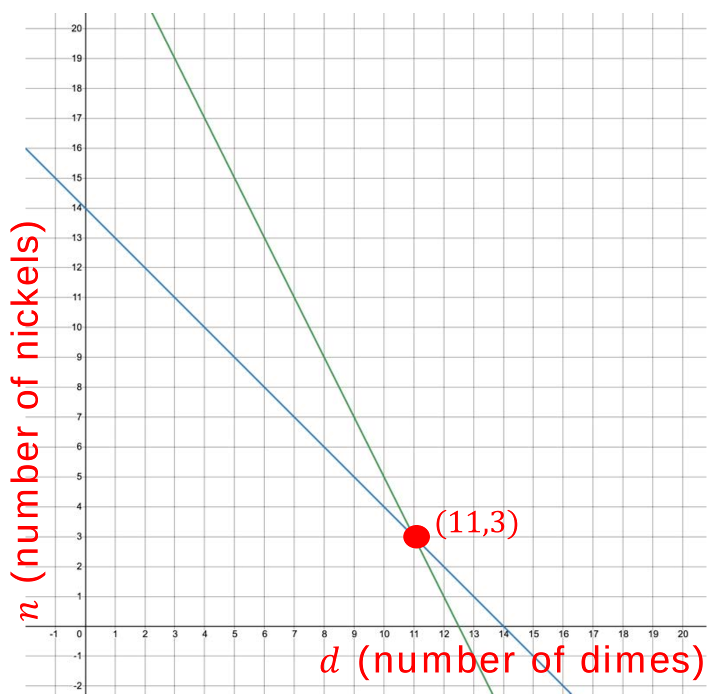
1. Graph the system of equations on the coordinate plane.

2. Complete the statement.

The point (__20__ , __14__) is the solution to the system.

Alexis has a total of 14 nickels and dimes worth $\$1.25$. The system of equations $\begin{cases} n = 14 - d \\ n = 25 - 2d \end{cases}$ can be used to represent the amount of quarters, q, and the amount of dimes, d.

The graph of the system is shown.



3. Label the axes.

4. What are any possible constraints on the x-values?

$x \geq 0$

5. What are any possible constraints on the y-values?

$y \geq 0$

6. What is the solution to this system of equations?

(11,3)

7. Interpret the meaning of the solution in the context.

Alexis has 11 dimes and 3 nickels, totaling $1.25.

Chen went to the grocery store and bought apples for $\$1.50$ each and oranges for $\$0.60$ each. He bought a total of 7 pieces of fruit and spent $\$6$. The system of equations $\begin{cases} y = 7 - x \\ y = -\frac{2}{5}x + 4 \end{cases}$ can be used represents the situation where x is the number of oranges and y is the number of apples bought.

8. Graph the system of equations. Label each line with the fruit it represents.

9. What is the solution to the system?

(5,2)

10. Interpret the meaning of the solution in the context.

Chen bought 5 oranges and 2 apples at the grocery store.